Three structural properties of the stability relation: intersecting chords, unstable chords, and invariant blocks

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Editorial note. The counted statement is the author's labelled lemma 'lem:properties of core' (source lines 873-881); the proof environment begins on the next line after \end{lemma} apart from one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's definition of an m-invariant set, which the third part of the statement and its proof invoke.

Editorial note. The retained preamble is the editorially prepared standalone preamble retained for this source.

Editorial note. Every label used inside the retained text is defined inside the retained spans (the self-reference in the enumeration options is to the lemma's own label), so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: the structural properties of the stability relation for matchings (author label lem:properties of core).

Allowed standard prerequisites (editorial).

- Stable and unstable chords. A chord is stable if it belongs to the core, equivalently if its endpoints are not removed during the reduction process; otherwise it is unstable.. The three parts are statements about stability and are proved by tracking which chords can be removed.
- Invariant set and restricted matching. A set S is m-invariant if every chord of m meets S in either two or zero vertices, and res_S(m) is the restriction of m to S.. The third part assumes [i,j] is m-invariant and restricts the matching to it.
- Short chords and reduction. The reduction process repeatedly removes short chords, i.e. chords of the form (i,i+1), together with their endpoints.. Both the first and second parts argue about which chords can become short.

Definition 0.1. Given a matching $m \in \mathcal{M}_N$, we say that a set $S \subseteq [N]$ is m-invariant if for every chord $(i,j) \in m$ we have $i \in S$ if and only if $j \in S$. Moreover, for a matching m and an m-invariant set S we define the restriction of m to S, denoted $\mathrm{res}_S(m)$, to be a matching on S which is obtained by removing all the vertices not in S from m.

Lemma 0.2. The reduction process has the following properties:

(1) If a chord intersects another chord then it is stable.

(2) A chord (i,j) is stable if and only if there exists a stable vertex in $\{i+1,\dots,j-1\}$.

(3) Given a matching m and $i < j$, if the set $[i, j]$ is m-invariant (as defined in Definition 0.1) and the restricted matching $\mathrm{res}_{[i,j]}(m)$ is perfect and non-crossing, then $\ell \notin \mathrm{Stable}(m)$ for all $\ell \in [i, j]$.

Proof.

1. Assume that $(i_1, i_3),\ (i_2, i_4) \in m$ for $i_1 < i_2 < i_3 < i_4$. As long as the chord (i_2, i_4) is not removed, the chord (i_1, i_3) does not become short and cannot be removed. On the other hand, as long as the chord (i_1, i_3) is not removed, the chord (i_2, i_4) cannot be removed. Therefore, both chords cannot be removed during the reduction process.

2. If the chord (i, j) is unstable, then after removing some unstable vertices it becomes short, implying that every vertex between i and j is unstable. On the other hand, if all the vertices between i and j are unstable, then they will eventually be removed, making the chord (i, j) short, so the chord (i, j) is unstable too.

3. Since the set $[i, j]$ is m-invariant and the restricted matching is perfect, Definition 0.1 implies that for every $i \leq i_1 \leq j$ there exists $i \leq i_2 \leq j$ such that $i_1 \neq i_2$ and $(i_1, i_2) \in m$ or $(i_2, i_1) \in m$. Notice that the restricted matching $\mathrm{res}_S(m)$ where $S = \mathrm{Stable}(m) \cap [i, j]$ (i.e. the matching that consists of the stable chords $(i_1, i_2) \in m$ with $i \leq i_1 < i_2 \leq j$) is short-chord-free, non-crossing and perfect. The only such a matching is the empty matching $\emptyset \in \mathcal{M}_0$, so every $i \leq i_1 \leq j$ is unstable. □